

Equation (3.3.9) relates the global displacement matrix $\{d\}$ to the local displacement $\{d'\}$ as

$$\{d'\} = [T]\{d\} \quad (3.3.10)$$

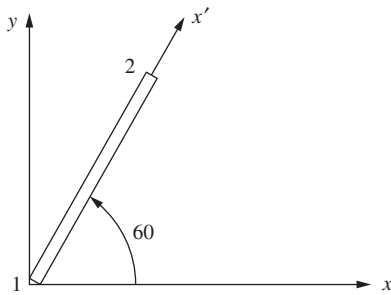
where

$$\{d\} = \begin{Bmatrix} u \\ v \end{Bmatrix}, \quad \{d'\} = \begin{Bmatrix} u' \\ v' \end{Bmatrix}, \quad [T] = \begin{bmatrix} C & S \\ -S & C \end{bmatrix} \quad (3.3.11)$$

The matrix $[T]$ is called the *transformation* (or *rotation*) *matrix*. For an additional description of this matrix, see Appendix A. It will be used in Section 3.4 to develop the global stiffness matrix for an arbitrarily oriented bar element and to transform global nodal displacements and forces to local ones.

EXAMPLE 3.2

The global nodal displacements at node 2 have been determined to be $u_2 = 0.1$ in. and $v_2 = 0.2$ in. for the bar element shown in Figure 3–8. Determine the local x displacement at node 2.



■ **Figure 3–8** Bar element with local axis x' acting along the element

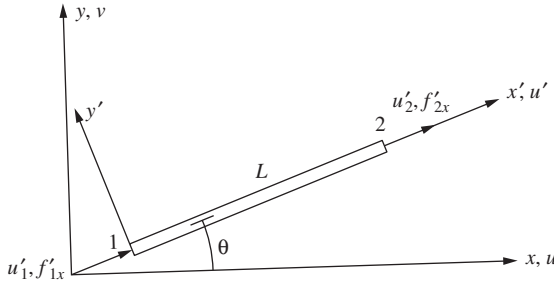
SOLUTION:

Using Eq. (3.3.5) for u' applied to node 2, we obtain

$$u'_2 = (\cos 60^\circ)(0.1) + (\sin 60^\circ)(0.2) = 0.233 \text{ in.}$$

3.4 Global Stiffness Matrix for Bar Arbitrarily Oriented in the Plane

We now consider a bar inclined at an angle θ from the global x axis identified by the local axis x' directed from node 1 to node 2 along the direction of the bar, as shown in Figure 3–9. Here positive angle θ is taken counterclockwise from x to x' .



■ **Figure 3-9** Bar element arbitrarily oriented in the global $x - y$ plane

We now use Eq. (3.1.13) where a prime notation is used to denote the local element stiffness matrix $\{k'\}$ which relates the local coordinate nodal forces $\{f'\}$ to local nodal displacements $\{d'\}$ as shown by Eq. (3.4.1).

$$\begin{Bmatrix} f'_{1x} \\ f'_{2x} \end{Bmatrix} = \frac{AE}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} u'_1 \\ u'_2 \end{Bmatrix} \quad (3.4.1)$$

or

$$\{f'\} = [k']\{d'\} \quad (3.4.2)$$

We now want to relate the global element nodal forces $\{f\}$ to the global nodal displacements $\{d\}$ for a bar element arbitrarily oriented with respect to the global axes as shown in Figure 3-9. This relationship will yield the global stiffness matrix $[k]$ of the element. That is, we want to find a matrix $[k]$ such that

$$\begin{Bmatrix} f_{1x} \\ f_{1y} \\ f_{2x} \\ f_{2y} \end{Bmatrix} = [k] \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} \quad (3.4.3)$$

or, in simplified matrix form, Eq. (3.4.3) becomes

$$\{f\} = [k]\{d\} \quad (3.4.4)$$

We observe from Eq. (3.4.3) that a total of four components of force and four of displacement arise when global coordinates are used. However, a total of two components of force and two of displacement appear for the local-coordinate representation of a spring or a bar, as shown by Eq. (3.4.1). By using relationships between local and global force components and between local and global displacement components, we will be able to obtain the global stiffness matrix. We know from transformation relationship Eq. (3.3.5) that

$$\begin{aligned} u'_1 &= u_1 \cos \theta + v_1 \sin \theta \\ u'_2 &= u_2 \cos \theta + v_2 \sin \theta \end{aligned} \quad (3.4.5)$$

In matrix form, Eqs. (3.4.5) can be written as

$$\begin{Bmatrix} u'_1 \\ u'_2 \end{Bmatrix} = \begin{bmatrix} C & S & 0 & 0 \\ 0 & 0 & C & S \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} \quad (3.4.6)$$

or as

$$\{d'\} = [T^*]\{d\} \quad (3.4.7)$$

where

$$[T^*] = \begin{bmatrix} C & S & 0 & 0 \\ 0 & 0 & C & S \end{bmatrix} \quad (3.4.8)$$

Similarly, because forces transform in the same manner as displacements, we replace local and global displacements in Eq. (3.4.6) with local and global forces and obtain

$$\begin{Bmatrix} f'_{1x} \\ f'_{2x} \end{Bmatrix} = \begin{bmatrix} C & S & 0 & 0 \\ 0 & 0 & C & S \end{bmatrix} \begin{Bmatrix} f_{1x} \\ f_{1y} \\ f_{2x} \\ f_{2y} \end{Bmatrix} \quad (3.4.9)$$

Similar to Eq. (3.4.7), we can write Eq. (3.4.9) as

$$\{f'\} = [T^*]\{f\} \quad (3.4.10)$$

Now, substituting Eq. (3.4.7) into Eq. (3.4.2), we obtain

$$\{f'\} = [k'][T^*]\{d\} \quad (3.4.11)$$

and using Eq. (3.4.10) in Eq. (3.4.11) yields

$$[T^*]\{f\} = [k'][T^*]\{d\} \quad (3.4.12)$$

However, to write the final expression relating global nodal forces to global nodal displacements for an element, we must invert $[T^*]$ in Eq. (3.4.12). This is not immediately possible because $[T^*]$ is not a square matrix. Therefore, we must expand $\{d'\}$, $\{f'\}$, and $[k']$ to the order that is consistent with the use of global coordinates even though f'_{1y} and v'_{2y} are zero. Using Eq. (3.3.9) for each nodal displacement, we thus obtain

$$\begin{Bmatrix} u'_1 \\ v'_1 \\ u'_2 \\ v'_2 \end{Bmatrix} = \begin{bmatrix} C & S & 0 & 0 \\ -S & C & 0 & 0 \\ 0 & 0 & C & S \\ 0 & 0 & -S & C \end{bmatrix} \begin{Bmatrix} u_1 \\ v_1 \\ u_2 \\ v_2 \end{Bmatrix} \quad (3.4.13)$$

$$\{d'\} = [T]\{d\} \quad (3.4.14)$$

where

$$[T] = \begin{bmatrix} C & S & 0 & 0 \\ -S & C & 0 & 0 \\ 0 & 0 & C & S \\ 0 & 0 & -S & C \end{bmatrix} \quad (3.4.15)$$

Similarly, we can write

$$\{f'\} = [T]\{f\} \quad (3.4.16)$$

because forces are like displacements—both are vectors. Also, $[k']$ must be expanded to a 4×4 matrix. Therefore, Eq. (3.4.1) in expanded form becomes

$$\begin{Bmatrix} f'_{1x} \\ f'_{1y} \\ f'_{2x} \\ f'_{2y} \end{Bmatrix} = \frac{AE}{L} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} u'_1 \\ v'_1 \\ u'_2 \\ v'_2 \end{Bmatrix} \quad (3.4.17)$$

In Eq. (3.4.17), because f'_{1y} and f'_{2y} are zero, rows of zeros corresponding to the row numbers f'_{1y} and f'_{2y} appear in $[k']$. Now, using Eqs. (3.4.14) and (3.4.16) in Eq. (3.4.2), we obtain

$$[T]\{f\} = [k'][T]\{d\} \quad (3.4.18)$$

Equation (3.4.18) is Eq. (3.4.12) expanded. Premultiplying both sides of Eq. (3.4.18) by $[T]^{-1}$, we have

$$\{f\} = [T]^{-1}[k'][T]\{d\} \quad (3.4.19)$$

where $[T]^{-1}$ is the *inverse* of $[T]$. However, it can be shown (see Problem 3.28) that

$$[T]^{-1} = [T]^T \quad (3.4.20)$$

where $[T]^T$ is the *transpose* of $[T]$. The property of square matrices such as $[T]$ given by Eq. (3.4.20) defines $[T]$ to be an orthogonal matrix. For more about orthogonal matrices, see Appendix A. The transformation matrix $[T]$ between rectangular coordinate frames is orthogonal. This property of $[T]$ is used throughout this text. Substituting Eq. (3.4.20) into Eq. (3.4.19), we obtain

$$\{f\} = [T]^T[k'][T]\{d\} \quad (3.4.21)$$

Equating Eqs. (3.4.4) and (3.4.21), we obtain the global stiffness matrix for an element as

$$[k] = [T]^T[k'][T] \quad (3.4.22)$$

Substituting Eq. (3.4.15) for $[T]$ and the expanded form of $[k']$ given in Eq. (3.4.17) into Eq. (3.4.22), we obtain $[k]$ given in explicit form by

$$[k] = \frac{AE}{L} \begin{bmatrix} C^2 & CS & -C^2 & -CS \\ & S^2 & -CS & -S^2 \\ & & C^2 & CS \\ \text{Symmetry} & & & S^2 \end{bmatrix} \quad (3.4.23)$$

Equation (3.4.23) is the explicit stiffness matrix for a bar arbitrarily oriented in the $x-y$ plane.

Now, because the trial displacement function Eq. (3.2.6) and Figure 3-5 was assumed piece-wise-continuous element by element, the stiffness matrix for each element can be summed by using the direct stiffness method to obtain

$$\sum_{e=1}^N [k^{(e)}] = [K] \quad (3.4.24)$$

where $[K]$ is the total stiffness matrix and N is the total number of elements. Similarly, each element global nodal force matrix can be summed such that

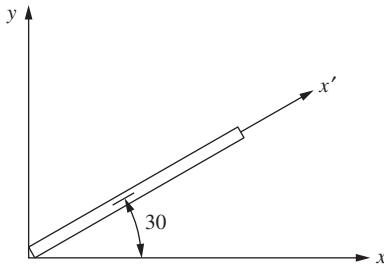
$$\sum_{e=1}^N \{f^{(e)}\} = \{F\} \quad (3.4.25)$$

$[K]$ now relates the global nodal forces $\{F\}$ to the global nodal displacements $\{d\}$ for the whole structure by

$$\{F\} = [K]\{d\} \quad (3.4.26)$$

EXAMPLE 3.3

For the bar element shown in Figure 3-10, evaluate the global stiffness matrix with respect to the $x-y$ coordinate system. Let the bar's cross-sectional area equal 2 in.^2 , length equal 60 in. , and modulus of elasticity equal $30 \times 10^6 \text{ psi}$. The angle the bar makes with the x axis is 30° .



■ Figure 3-10 Bar element for stiffness matrix evaluation

SOLUTION:

To evaluate the global stiffness matrix $[k]$ for a bar, we use Eq. (3.4.23) with angle θ defined to be positive when measured counterclockwise from x to x' . Therefore,